

ABHINAV ARORA

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EDUCATION

California State University - Fullerton

Bachelors of Science in Computer Science

Fullerton, California

SUMMARY

Abhinav Arora is an agile AI engineer, specializing in Machine Learning, PHP, Javascript, SQL and Python with proven ability to optimize the performance of back-end infrastructure. A respected industry track record of effectively diagnosing and resolving complex database operations, and managing web-Servers hosted on cloud environments. With years of experience in production debugging and providing real-time customer solutions.

TECHNICAL SKILLS

- Javascript, Python, Laravel, Oracle, SQL, AWS, Azure, Terraform, Git, JIRA, R, Go, Rust, Django, GraphQL, C++, Java, Qlik, Jenkins, Docker, Swift.

EXPERIENCE

Artificial Intelligence Engineer | New Hampshire Ball Bearings, Inc. | Los Angeles, California

March 2025 - Present

- Built secure, offline RAG chatbots using LLaMa 3.2 LLM to help each cell quickly access and query standard procedures and documents.
- Optimized inference latency using model compression techniques like quantization, cutting compute costs by 60% with no accuracy loss.
- Automated 3-Way Match process using fine-tuned Mistral-7B model, improving accuracy by almost 70% and reducing manual workload.
- Engineered data pipelines & warehouses to support AI inferencing, cutting reporting time from hours to seconds.
- Deployed Mixtral vision model for AI inferencing to extract line-item data from unstructured Purchase orders, streamlining ERP ingestion.
- Built and fine-tuned CNNs, RNNs, and Transformers in TensorFlow for classification and detection using transfer learning and data augmentation.

Software Engineer | Bega North America | Carpinteria, California

June 2022 - March 2025

- Automated order entry using Amazon Bedrock's Generative AI, reducing manual data entry by 90% and boosting order processing speed.
- Led migration of legacy Qlik on-premise infrastructure to Qlik Cloud and implemented real-time data warehousing solution on AWS Redshift.
- Boosted system scalability by redesigning architecture with microservices, using Kubernetes for orchestration and Docker for containerization.
- Spearheaded the integration of AI and machine learning chatbots into Qlik Dashboards, enhancing data functionality and user experience.
- Integrated GraphQL with PHP and React, optimizing API performance and reducing redundant API calls by 60%, leading to faster UI rendering.
- Developed an EDI portal on AWS EC2 with RDS, automating the 3-way matching process and increasing transaction efficiency by 15%.
- Implemented custom C++ drivers and middleware for seamless communication between warehouse conveyor belts and backend order systems.
- Refactored monolithic PHP applications into a modern MVC structure using Laravel, improving maintainability and reducing technical debt.
- Wrote multi-threaded C++ applications to handle parallel processing for barcode scanning and sorting, reducing processing time by 50%.
- Designed modular and reusable components for backend services, and clean architecture and optimized performance for AWS deployments.

Software Engineering Intern | Solar Insure Inc. | Costa Mesa, California

January 2022 - May 2022

- Developed and integrated RESTful APIs using Flask and PHP, reducing data retrieval times by 30% and enhancing efficiency of the web portal.
- Architected multi-region deployments of web applications, ensuring high availability and disaster recovery readiness for customers.
- Developed a secure and efficient payment processing system by integrating the VISA API to ensure reliable transaction handling of claims.
- Integrated Salesforce API to retrieve and display customer claims data, enabling real-time synchronization between Salesforce and web portal.
- Automated unit and integration testing using PyTest and Postman, achieving 95% test coverage for critical AWS-hosted services.

Javascript Developer | California State University Fullerton | Fullerton, California

September 2021 - January 2022

- Developed a custom PHP web-portal with a React frontend, integrating AI/ML virtual agent-mentors for a University Career Fair chatbot.
- Integrated AJAX to load content dynamically without refreshing the page, reducing server load by 20% and enhancing user experience.
- Implemented Redux for efficient state management, reducing bugs and inconsistencies by 30% and enhancing application stability.
- Automated real-time data fetching and updates using React hooks and WebSockets, reducing data latency by 60% and improving UI/UX scores.
- Conducted a thorough assessment of the SQL Server database performance using monitoring tools and performance analyzers like SQL Sentry.

Information Technology Intern | Sensorex | Garden Grove, California

August 2021 - December 2021

- Engineered a new invoice processing feature using Laravel and React, cutting processing times by 40% and improving financial operations.
- Implemented user authentication and authorization with Flask-Security and AJAX, ensuring secure access and data protection.
- Developed real-time transaction tracking dashboards using React, providing finance teams with instant visibility into payout statuses.
- Optimized Laravel backend queries and introduced Redis caching, resulting in a 50% decrease in database query execution times.

Student Software Developer | California State University Fullerton | Fullerton, California

March 2021 - May 2022

- Debugged production bugs, and deployed live fixes on Linux servers, to maintain a 24/7 University IT infrastructure.
- Secured SQL Server Linux instances by configuring firewalls and IP whitelisting, reducing unauthorized access attempts by 30%.

Research Student Assistant | California State University Fullerton | Fullerton, California

June 2019 - August 2020

- Enhanced data models and algorithms for efficient SQL queries, improving data retrieval and analysis from the Azure Data Lake.
- Revamped ETL processes using SSIS to extract, transform, load data into the analytical data models, and detailed logging of execution metrics.

Information Technology Student Assistant | California State University Fullerton | Fullerton, California

May 2019 - August 2019

- Utilized PHPUnit for comprehensive unit testing on PHP web servers, achieving 85% test coverage and reducing frontend bugs by 40%.
- Designed standard operating procedures for database administration and troubleshooting processes to ensure continuous knowledge transfer.

BOARD POSITION

ACM External Vice President

May 2020 - May 2021

- Organized networking events with senior management alumni to build industry connections and strengthen professional relationships.

CERTIFICATIONS

- IBM - Enterprise Design Thinking Practitioner.

August 2021-Present

PERSONAL PROJECTS

Aurora

November 2022- March 2024

- Developed and deployed a cloud-native venture trading capital portfolio management application using Python and Django, hosted on AWS EC2.

Zero-Trust Application Access Platform

October 2022- September 2023

- Created a Docker container remote access platform using Zero-Trust Security framework and Network Hole punching.